Read the Piot engine as an invention

Task 3.1 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Use Fig. 1 of Thomas Piot's 1891 UK patent 20,081 and the teacher demonstration. Identify the heat source, vessel, tubes and surrounding water. Trace the route with arrows.

Open Piot’s original patent and figure (also linked in the lesson)

Trace the engine tubes and mark the waterline on a copy you draw. Pick a fixed travel distance for later tests.

Feature	Purpose inferred	Evidence in figure or demonstration
Vessel		
Tube opening		
Heat source		

02 Plan it on paper

Identify the existing design, the function to preserve, and one measurable improvement. Label each separately.

My prediction, chosen feature or design limit

Read the Piot engine as an invention

Task 3.1 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Observe a teacher-run demonstration. Sketch the motion and identify which explanations are observations and which are inferences. Swap annotated sketches. Have a partner trace water into and out of the mechanism and identify one part needing inspection.

What you saw	What you infer	How to check it
Motion		
Water path		
Possible change		

04 Use the reading

Ch. 17 | In Deep | printed pp. 155-165
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

What would count as an improvement on this patent? Give a metric and the starting conditions you will keep.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Inspect a baseline boat at each station

Task 3.2 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Copy the dimensions from the approved physical pattern. Mark cut and fold references on your sketch. The teacher handles designated sharp or heated operations. Record cure time from the actual product label.

Take cut, fold and opening dimensions from the teacher-approved station pattern or pilot sample. Record the units and check the paired measurements before forming.

Part or check	Pattern size / units	Measured size	Inspection result
Hull			
Chamber edges			
Tube position			
Symmetry			

02 Plan it on paper

Record the demonstrated station sequence. Mark which steps require an adult, inspection or curing time.

My prediction, chosen feature or design limit

Inspect a baseline boat at each station

Task 3.2 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

At supervised stations, follow the teacher-piloted sequence for the baseline hull and engine. Label and store parts for the adhesive cure time posted at the station. Inspect alignment, sealed seams, tube clearance, and hull symmetry. Have the teacher sign off before a powered basin test.

Station	Operation or check	Time / cure / initials
Dry fit		
Seal / cure		
Leak inspection		
Teacher release		

04 Use the reading

Ch. 17 | In Deep | printed pp. 155-165

Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Explain one quality check that must happen before the next assembly step. Locate it on your drawing.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Establish a boat and fluid baseline

Task 3.2–3.3 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Record the start rule and load. Speed = distance ÷ elapsed time. At the syringe station, mark initial piston positions; compare equal input strokes with air and water.

Mark start and finish lines. Record distance, time, load and test conditions for three comparable runs.

Boat run	Distance and units	Time (s)	Speed / load
1			
2			
3			

02 Plan it on paper

Predict what changes when linked syringes contain air instead of water. Mark the initial piston positions.

My prediction, chosen feature or design limit

Establish a boat and fluid baseline

Task 3.2–3.3 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Run the approved baseline boat at the teacher station. At the fluid station move the input piston a fixed distance and measure the output travel. Repeat each fluid trial. Look for bubbles, slipping tubing, or a changed load before interpreting differences.

Fluid and test	Input travel	Output travel	Units / leaks or bubbles
Air 1			
Air 2			
Water 1			
Water 2			

04 Use the reading

Ch. 18 | Shooting for the Moon | printed pp. 167-176
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Give one baseline boat speed or a syringe travel comparison. Explain what you held constant and why.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Use two pistons to move a load

Task 3.4 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Sketch the two syringes, their sizes, the connected tube and the load. Predict travel when the sizes swap. Keep sample and input travel constant.

Measure input and output piston travel with the same reference edge. Record syringe sizes.

Arrangement	Input size	Output size	Load / sample
A			
B swapped			

02 Plan it on paper

Draw the input piston, tube, output piston, and load. Predict the effect of swapping the syringe sizes.

My prediction, chosen feature or design limit

Use two pistons to move a load

Task 3.4 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Mount the connected syringes so an input stroke drives the output piston against a soft sample. Keep hands clear. Swap roles to inspect the mechanism. Run three equal input strokes in both arrangements. Compare output travel and the change in the sample.

Trial	A input / output travel	B input / output travel	Sample change
1			
2			
3			

04 Use the reading

Ch. 19 | Fuel from the Fields | printed pp. 177-184
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Which arrangement met the job better? Include travel measurements and identify the trade-off.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Explain one cycle of the boat engine

Task 3.5 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Draw a four-frame loop. Label the chamber, heat, water and tube mouths. Distinguish inferred pressure change from visible motion. Show how cooler fluid can enter again.

Mark heat input, surrounding cooler water, and tube openings. Identify where the team observed water motion.

State	Heat transfer	Pressure change inferred	Flow direction
Warm-up			
Expansion			
Cooling			
Return			

02 Plan it on paper

Draw successive states of the working fluid. Predict the direction of flow at two points in the cycle.

My prediction, chosen feature or design limit

Explain one cycle of the boat engine

Task 3.5 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Watch the approved demonstration. Annotate the cycle with changes in temperature, pressure, and movement. Discuss which changes are inferred. Have a partner trace the entire cycle back to its starting state. Repair a missing energy transfer or reversed arrow.

Claim	Observation	Inference or open question
Water motion		
Repeating cycle		

04 Use the reading

Ch. 20 | An Ingenious Engine | printed pp. 185-190

Ch. 21 | Energy from the Earth | printed pp. 191-198

Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Use two stages in the cycle to explain how heating can produce repeated motion. Identify evidence and one inference.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Compare the flow through two routes

Task 3.6–3.7 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Sketch routes A and B. Change one variable, such as tube length. Use the same water level or drive. Flow rate = collected volume ÷ collection time.

Mark a common head or driving condition. Measure tube length, collection time, and collected volume.

Route	Tube length / bore	Drive condition	Fixed outlet position
A			
B			

02 Plan it on paper

Draw two routes differing in one variable. Predict which will deliver more water in equal time.

My prediction, chosen feature or design limit

Compare the flow through two routes

Task 3.6–3.7 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Run both routes from the same starting condition. Keep tube ends and water level consistent. Collect three trials. Calculate volume/time with units. Check for kinks or changing head. Compare the direction of an outlet jet with the expected reaction.

Trial	A volume / time	B volume / time	A and B flow rates with units
1			
2			
3			

04 Use the reading

Ch. 22 | Good Chemistry | printed pp. 199-207
Ch. 23 | Down the Pipes | printed pp. 209-218
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Which change would you recommend testing on the boat or press? Support your prediction with measured flow rates.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Change one feature of a boat or press

Task 3.8 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Draw the baseline and the revision side by side. Label the changed dimension, the target and everything the next test must hold constant.

Choose one dimension or operating condition you can change. Measure the initial value and set a target.

Feature	Original and units	Revised target	Predicted effect
Change			
Fixed load			
Fixed start rule			

02 Plan it on paper

Sketch the revision and predict how it will change the selected metric. Keep the other test conditions fixed.

My prediction, chosen feature or design limit

Change one feature of a boat or press

Task 3.8 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Build one change under the same equipment approvals. Inspect and allow cure time before retesting. On the press route change one linkage or syringe arrangement. Run an unpowered inspection or leak test first. Have a second person check that the original test can still be repeated.

Inspection	Result	Who checked / next action
Unpowered fit		
Leak or motion		
Cure if needed		

04 Use the reading

Ch. 23 | Down the Pipes | printed pp. 209-218
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

What did you change, what stayed fixed, and what numerical outcome would support your prediction?

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Retest the revised machine

Task 3.8 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Match the earlier distance or piston stroke, load and start rule. Keep failed trials in the record. Use the same measure and units for both versions.

Use the same course length or input stroke, same load, and same start rule. Collect at least three revised trials.

Run	Baseline result	Revised result	Conditions or failed trial
1			
2			
3			

02 Plan it on paper

Make a two-column table for baseline and revision. Write the units at the top.

My prediction, chosen feature or design limit

Retest the revised machine

Task 3.8 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Run the revised machine, recording every attempt. Calculate the same measure used for the baseline. Compare typical values and spread. If a trial fails, describe the cause you observed before excluding it.

Comparison	Baseline	Revision	Interpretation
Typical value			
Range			
Limitation			

04 Use the reading

Ch. 20 | An Ingenious Engine | printed pp. 185-190
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Does the data support your improvement claim? Include both values, units, and a limit of the comparison.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Defend a measured improvement

Task 3.9 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Draw the modified feature large enough to label. Use the same metric for both versions. Check that each teammate can locate the change and describe its purpose.

Check every labeled dimension and compare the claim with the recorded before-and-after results.

Claim component	Evidence, unit or labeled detail
Original need	
Modified feature	
Before / after	

02 Plan it on paper

Organize the presentation around the original need, change, mechanism, and evidence. Assign a speaking role to each team member.

My prediction, chosen feature or design limit

Defend a measured improvement

Task 3.9 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Demonstrate the working device using the approved setup. Give a peer the drawing and ask them to locate the modified feature. Respond to one challenge about the mechanism or test. Revise an overstrong claim using the actual measurements.

Review check	What reviewer found	What you clarified
Find the change		
Challenge a claim		
Next test		

04 Use the reading

Ch. 17 | In Deep | printed pp. 155-165
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Which evidence supports your claim, and which question would need another test?

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.